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(Phase-I)

On

Autonomous smart irrigation system using AI

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CANDIDATE'S DECLARATION

We hereby certify that the work presented in the **Phase-I report** of the project entitled "**Autonomous smart irrigation system using AI**", in partial fulfillment of the requirements for the degree of **Bachelor of Technology** in the **School of Computing Science and Engineering, IILM University**, Greater Noida, is an original work carried out by us under the supervision of **Mr. Shobhit Agrawal**, School of Computing Science and Engineering, IILM University, Greater Noida.

The work presented in this Phase-I report has not been submitted by us for the award of any other degree or qualification at this or any other institution.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

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ABSTRACT

Agriculture remains one of the largest consumers of freshwater, yet a significant portion of irrigation water is lost due to inefficient scheduling and poor decision-making, with estimates suggesting more than 40% of redirected irrigation water never reaches the crops it is intended for . Agriculture is one of the largest consumers of freshwater through irrigation. Irrigation is often wasteful, especially when done manually or scheduled poorly. Irrigation scheduling based on software can help optimize water consumption. Currently, solutions are available based on IoT devices or other hardware sensors to help monitor the soil and operate irrigation. Most solutions based on sensors tend to be very expensive and can be a prohibitive cost for small-scale farmers or for simple research and experimentation.A review of the literature suggests that current research into smart irrigation systems tends to focus on physical monitoring and controlling devices with limited attention given to optimizing irrigation based on a software system alone.This project aims to create a software-based irrigation decision support system, which will utilize machine learning algorithms on realistic and synthetically generated data to estimate the soil's moisture characteristics and irrigation requirements. The system will be developed to produce irrigation recommendations without requiring additional hardware to monitor the soil.This review revealed that while considerable work has been done on hardware-based monitoring and control systems, comparatively little attention has been given to fully software-driven, data-centric approaches the objective of this project is to design and develop a working software-based irrigation decision-making system that exploit real-world and synthetically generated datasets to predict soil moisture conditions and irrigation requirements using ml techniques, without depending on physical sensors or hardware devices.

1. Problem Definition

Irrigation is one of the most essential yet resource-intensive practices in agriculture. Water used for irrigation is drawn from a variety of sources — groundwater, surface water bodies, and even treated wastewater. Despite its importance, a large proportion of irrigation water is wasted before it ever reaches the crop. Studies estimate that more than 40% of redirected irrigation water is lost at the farm level through deep percolation, surface runoff, or simply inefficient timing , water that is applied either too early, too late, or in the wrong quantity relative to what the crop actually needs.Traditional irrigation still depends heavily on fixed schedules or the farmer's personal judgment.

On the other hand, most modern "smart" irrigation systems that attempt to solve this problem do so by introducing physical infrastructure: soil moisture sensors, IoT devices, wireless sensor networks, and communication hardware.

Sensor-based monitoring equipment remains expensive to purchase, install, and maintain, which makes such systems really difficult to small-scale farmers and impractical for many research and academic settings where hardware, deployment, and field testing are simply not achievable.

It aims to explore whether irrigation scheduling and water-requirement prediction can be handled effectively through a fully software based and data driven approach using machine learning and models trained on real and synthetically created data .

Research Gap

So from all the reviewed literature , there is this clear and consistent pattern that we have analyzed and i.e nearly all existing smart irrigation systems regardless of ml technique used like random forest , SVM, KNN, Deep Belief Networks, Neural Networks or other models are technically hardware based or hardware dependent. Relying on physical soil moisture sensors, microcontrollers such as Arduino,ESP32,STM32, Rasberry Pi and often continuous

use of internet or cloud connectivity for data handling and decision making . This create significant barriers like high deployment cost, much dependency on network or internet connectivity , poor generalization across different type of crops, soil, and climates. Most models are trained on narrow site specific datasets and limited scalability for small scale farmers or research environment where physical deployment is not practical. Various study noted these limitations , the need for larger and more vast large amount of datasets, retraining the requirements for new crop, climatic conditions.

Point to be noted that none of the reviewed works proposed a purely software based , hardware independent smart irrigation decision system that is trained on a combination of real and synthetically generated data to overcome dataset scarcity and other issues. This gap forms the core motivation for the present project.

2. Literature Survey

A review of thirty research papers relevant to AI-driven and IoT-based smart irrigation was carried out to understand existing techniques, datasets, evaluation metrics, and reported outcomes. The reviewed works range from hardware- integrated prototypes to systematic literature reviews, and collectively cover classical Machine Learning classifiers (Decision Tree, Random Forest, SVM, KNN, Naive Bays), regression and time-series approaches (Linear Regression, PLSR, LSTM, XGBoost), deep learning and reinforcement learning methods (CNN, Deep Belief Networks, Q-Learning), and edge/IoT deployment strategies (ESP32, Raspberry Pi, STM32, TinyML). A summary of the reviewed literature is presented below.

S. No.	Title	Author(s)/Year	Technique/Method	Key Result
1	Smart Irrigation Powered by Artificial Intelligence (AI)	Harini S; Shobana A; Sapna S; Vedhashree R; Prabhakaran M — 2024	AI/ML-based irrigation planning; Arduino Uno; DHT11; soil-humidity sensor; solenoid valves; Blynk IoT monitoring.	Authors report potential savings of up to 50% in operational costs; system supports remote monitoring and automatic water supply control.
2	IoT with Evolutionary Algorithm Based Deep Learning for Smart Irrigation System	P. Suresh; R. H. Aswathy; Sridevi Arumugam; Amani Abdulrahman Albraikan; Fahd N. Al-Wesabi; Anwer Mustafa Hilal; Mohammad Alamgeer — 2023	IoT sensors; Arduino; DSVM-based regression; clustering; Artificial Immune Optimization Algorithm (AIOA); Deep Belief Network (DBN); cloud processing.	Reported average accuracy 97.1%, precision 98.0%, recall 98.1% and F1-score 98.0%; the paper reports higher performance than the compared baseline models.
3	Intelligent Irrigation System With Smart Farming Using ML and Artificial Intelligence Techniques	P. Manikandan; S. Saravanan; C. Nagarajan — 2024	Ensemble SVM-KNN; IoT sensors; cloud service/data storage; web application for visualization and monitoring.	Ensemble SVM-KNN achieved 90% accuracy, outperforming individual SVM and KNN (72% and 76%).

4	Smart Irrigation System Using Artificial Intelligence and IOT	C.G. Anu Chandra; Dinesh Kumar S; Mohammad Aousaf; Sumitra Sharma Phurailatpam — 2025	AI-powered decision support; regression/predictive analytics; weather-aware processing; cloud platform; automated valve/pump control.	Prototype maintained stable moisture predictions across soil/weather conditions; controlled evaluation reported significantly lower water use than manual irrigation, but no exact percentage was stated.
5	Design and Implementation of an Integrated IoT and Artificial Intelligence System for Smart Irrigation Management	Taoufik Benhmad; Chibani Belgacem Rhaimi; Saleh Alomari; Leenah Aljuhani — 2024	AI prediction/regression; IoT sensor network; cloud-based wireless communication; remote monitoring/control.	Reported substantial water reduction, improved crop health and yield; the paper reports up to 35% water reduction and 25% yield increase in its results discussion.
6	Autonomous Crop Irrigation System using Artificial Intelligence	Savita Choudhary; Vipul Gaurav; Abhijeet Singh; Susmit Agarwal — 2019	Random Forest Classification/Regression for rainfall; ARIMA for rainfall time series; PLSR for crop water requirement; Raspberry Pi + NodeMCU/IoT control.	Controlled-environment testing reported 80% overall accuracy. Punjab annual rainfall prediction: 545 mm vs 549.5 mm, reported 83% accuracy. Pump is automatically controlled based on predicted rain and soil moisture.
7	AI-Driven Smart Irrigation System Using Edge-Based Embedded Controllers	Fahad Al-Jame; Perera Manthila — 2026	Lightweight ML/TinyML; Raspberry Pi 4 for AI inference; STM32 for actuation; local edge decision-making using environmental and soil-moisture data.	Decision accuracy >94%; inference time 34 ms; water use reduced up to 38%; crop yield increased up to 20%; power consumption <2.5 W.
8	Implementation of Smart Irrigation using IoT and Artificial Intelligence	Tace Y.; Elfilali S.; Tabaa M.; Leghris C. — 2023	KNN, SVM and Naive Bayes; Arduino sensor acquisition; Node-RED monitoring/storage; MongoDB Cloud; supervised learning with pump-decision labels.	KNN achieved 98.4% accuracy and 0.016 RMSE among the compared models.

9	Smart Irrigation System Based on IoT and Machine Learning	Y. Tace; M. Tabaa; S. El→lali; C. Leghris; H. Bensag; E. Renault — 2022	Soil-moisture, temperature, air-humidity and rain sensing; Node-RED; MongoDB; KNN; Neural Network; Naïve Bayes; SVM; Logistic Regression; web application.	KNN achieved 98.3% accuracy with RMSE 0.12, ahead of the evaluated Neural Network, Naïve Bayes, SVM and Logistic Regression models in the reported experiment.
10	A Smart Irrigation Control System That Leverages Machine Learning	M. Karthikaa; J. Steffib — 2025	IoT temperature + soil-moisture sensors with ESP32; Random Forest classifier for wet/dry soil-status prediction; SMOTE, ADASYN and Mix-up for class-imbalance handling; cross-validation; automatic pump ON/OFF control; FastAPI backend Edge-based TinyML/supervised for live prediction/control.	Random Forest baseline: 84% accuracy, 82% precision, 80% recall, 81% F1. SMOTE: 89%, 87%, 88%, 87%. ADASYN: 91%, 90%, 89%, 89%. Mix-up: 92% accuracy, 91% precision, 92% recall, 91% F1 — best result.

3. Objectives

Based on the problem, the lack of accessible, hardware-independent smart irrigation solutions and this project sets out the following objectives:

3.1. To collect and prepare suitable agricultural data Gather real-world datasets related to soil moisture, weather conditions, crop water requirements, and irrigation patterns from publicly available sources, and supplement these with synthetically generated datasets to address gaps, different cases, and scenarios not sufficiently covered by real data.

3.2 To conduct a thorough review of existing approaches Study and compare existing smart irrigation techniques using particularly AI/ML based scheduling methods to identify best modeling approaches, common exceptions, and areas that remain under-explored, especially in the context of software- based only implementations.

3.3 Build a Prediction Model To develop a machine learning model that can predict the soil moisture and the amount of water needed by the crop. The model will leverage available agricultural data for predictions.

4. Scope

The main goal of this project is to create a software system that can use farming data to recommend when watering plants might be needed. The system will work through software and data. It will not use any sensors, internet of things devices or any physical equipment. Follow as:

Data Collection and Preparation

The project will use farming data that's already available. This data will be about soil, weather, crops and other farming activities. The data that is collected will be. Prepared before it is used for the model. If there is not real data some fake data may also be made for testing and showing how the system works.

Study of Existing Work

Some existing systems for watering plants and related research papers will be looked at. This will help to see how watering decisions are made now. It will also help to pick the machine learning methods for the new system.

Predictive Model

A machine learning model will be made using the farming data. The model will try to predict things that can help decide if watering is needed. These things include soil moisture and weather conditions.

Irrigation Decision-Making

The results, from the model will be used to give watering advice. For example the system might say when watering is needed and how to use resources when the plants need water. These decisions will be shown through the software. Will not control any real watering equipment.

Testing and Evaluation

The model will be checked using testing methods. The predictions will be compared with data when possible. This will help to see if the system is giving reliable watering advice

5. Feasibility Study

A feasibility study was conducted to ensure the project's viability based on the given timeframe, resources, and skill set of the team.

5.1-Technical Feasibility

The project is purely software-based, which means that there is no need for additional sensors, machinery, or other hardware components. It can be completed with the use of a computer or a laptop.

Programming and ML Tools

The programming language that will be used for the development of the project is Python, which includes such essential libraries as Pandas, NumPy, and Scimkit-Iearn. All of these modules are open-source and freely available on the internet.

Computer Requirements

The suggested project does not require any advanced PC hardware. A standard laptop or computer will be sufficient for running the code and training ML models. In case of insufficient performance, one might use Google Colab to access additional hardware resources for coding.

Team Skills

The project will require knowledge of Python programming, data processing, ML, and model training and validation. However, all of these skills are rather simple and can be obtained by the team members during the project's development.

5.2 Data Feasibility

The project will require information on soil, weather, crops, and irrigation. Such data can be found in multiple databases on the web, which will allow for the proper training and validation of the created model.

The availability of the required data can be considered a challenge; thus, there is an option to utilize synthetic data for testing purposes. This way, the researchers will be able to train the model with the use of fabricated information and later validate and test it with real-world data.

5.3 Overall Assessment

From this information, it can be concluded that the suggested project is completely viable and can be delivered within the given timeframe and resources. Firstly, it does not require any advanced or specialized hardware components beyond a standard computer or a laptop. Secondly, the necessary software tools and programming language, Python, are easily accessible and free to use. Finally, the project's data feasibility is not an issue as well due to the possibility of utilizing open-source datasets and synthetic data. Thus, the project can be developed and trained within the given timeframe and will successfully fulfill its stated goals and objectives.

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PLAGIARISM REPORT

The plagiarism/similarity-index report generated for this Phase-I report (e.g., via Turnitin or an equivalent institutionally approved tool) is to be attached here as a separate page prior to final submission, as required by the School of Computer Science and Engineering, IILM University, Greater Noida.